

Tao2.5

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1 Periodic Functions

Let $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Then f is bounded.

Proof: Consider the restriction $f : I \rightarrow \mathbb{C}$. Note that $f(I) = f(\mathbb{R})$. Since I is compact, so is $f(I)$, so $f(I)$ is bounded.

Suppose $f, g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$, $c \in \mathbb{R}$. Then

$$(f + g)(x + 1) = f(x + 1) + g(x + 1) = f(x) + g(x) = (f + g)(x),$$

$$(cf)(x + 1) = cf(x + 1) = cf(x),$$

$$(fg)(x + 1) = f(x + 1)g(x + 1) = f(x)g(x) = (fg)(x),$$

so $f + g, cf, fg \in C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$.

Suppose (f_n) converge uniformly to f and $f_n \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Then $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$.

Proof: Since $f_n \in C(\mathbb{R}, \mathbb{C})$, $f \in C(\mathbb{R}, \mathbb{C})$ as well. We have the point-wise limit

$$f(x + 1) = \lim_{n \rightarrow \infty} f_n(x + 1) = \lim_{n \rightarrow \infty} f_n(x) = f(x).$$

Let $x \in \mathbb{R}$. There exists a unique $n \in \mathbb{Z}$ and $a \in [0, 1)$ such that $x = n + a$.

Consider the set $Z = \{z \in \mathbb{Z} : z \leq x\}$. Note $Z \neq \emptyset$ by Archimedean principle, and is bounded above, so let $n = \sup Z$. Note that Z is closed since Z inherits the discrete topology from $\mathbb{R}$, so $n \in Z$. Hence, $n \leq x$. Let $a = x - n \geq 0$. Suppose FTSOC $a \geq 1$. Then $x - (n + 1) = (x - n) - 1 = a - 1 \geq 0$, so $n + 1 \in Z$, absurd. Hence, the existence of such a representation is shown.

Suppose $x = n + a = n' + a'$ for $n, n' \in \mathbb{Z}$ and $a, a' \in [0, 1)$. Then $n - n' = a' - a$. Sps FTSOC $n \neq n'$ (since otherwise $a = a'$). Then $|LHS| \geq 1$. However, $RHS \in (-1, 1)$, absurd.

$(C(\mathbb{R}/\mathbb{Z} : \mathbb{R}), d_\infty)$ is a submetric of $(C(\mathbb{R} : \mathbb{R}); d_\infty)$. Suppose (f_n) is Cauchy in $C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$. Since $C(\mathbb{R} : \mathbb{R})$ is complete, $f_n \rightarrow f$ for some $f \in C(\mathbb{R} : \mathbb{R})$. The previous part shows that $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$ as well, so $C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$ is complete.

2 Inner Products of Periodic Functions

Sps $f : I \rightarrow \mathbb{C}$ is bounded continuous. Define $\int_I f = \int_I \Re(f) + i \int_I \Im(f)$. $\int_I f$ is well-defined since $\Re(f), \Im(f)$ are also bounded continuous. Let $z = a + bi$. Then zf is also RI on I and $\int_I zf = z \int_I f$.

Proof: zf is bounded continuous.

$$\begin{aligned} \int_I zf &= \int_I (a + bi)(\Re(f) + i\Im(f)) = \int_I (a\Re(f) - b\Im(f)) + (b\Re(f) + a\Im(f))i \\ &= \int_I (a\Re(f) - b\Im(f)) + i \int_I (b\Re(f) + a\Im(f)) = a \int_I \Re(f) - b \int_I \Im(f) + bi \int_I \Re(f) + ai \int_I \Im(f) \\ &= a \left(\int_I \Re(f) + i \int_I \Im(f) \right) + bi \left(\int_I \Re(f) + i \int_I \Im(f) \right) \\ &= a \int_I f + bi \int_I f = (a + bi) \int_I f = z \int_I f. \end{aligned}$$

Suppose $f, g : I \rightarrow \mathbb{R}$ are bounded continuous. Then $\int_I (f + g) = \int_I f + \int_I g$.

Proof: $f + g$ is bounded continuous.

$$\begin{aligned} \int_I (f + g) &= \int_I \Re(f + g) + i \int_I \Im(f + g) \\ &= \int_I \Re(f) + \Re(g) + i \int_I \Im(f) + \Im(g) \\ &= \left(\int_I \Re(f) + i \int_I \Im(f) \right) + \left(\int_I \Re(g) + i \int_I \Im(g) \right) \\ &= \int_I f + \int_I g. \end{aligned}$$

Suppose $f : I \rightarrow \mathbb{R}$ is bounded continuous. Then so is $\bar{f} : I \rightarrow \mathbb{R}$ and $\int_I \bar{f} = \overline{\int_I f}$.

Proof:

$$\begin{aligned} \int_I \bar{f} &= \int_I \Re(f) - i \Im(f) = \int_I \Re(f) - i \int_I \Im(f) \\ &= \overline{\int_I \Re(f) + i \int_I \Im(f)} = \overline{\int_I f}. \end{aligned}$$

Suppose $f, g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$. Define $\langle f, g \rangle = \int_I f \bar{g}$. We have

$$\overline{\langle f, g \rangle} = \overline{\int_I f \bar{g}} = \int_I \bar{f} g = \int_I g \bar{f} = \langle g, f \rangle.$$

$$\langle f + g, h \rangle = \int_I (f + g) \bar{h} = \int_I f \bar{h} + \int_I g \bar{h} = \langle f, h \rangle + \langle g, h \rangle.$$

$$\langle zf, h \rangle = \int_I z f \bar{h} = z \int_I f \bar{h} = z \langle f, h \rangle.$$

$$\langle f, g + h \rangle = \int_I f \cdot \overline{(g + h)} = \langle f, g \rangle + \langle f, h \rangle.$$

$$\langle f, zg \rangle = \int_I f \overline{zg} = \bar{z} \langle f, g \rangle.$$

$$\langle 1, e^{2\pi i x} \rangle = \int_I e^{-2\pi i x} = \int_I \cos 2\pi x - i \sin 2\pi x = 0.$$

Note

$$\langle f, f \rangle = \int_I f \bar{f} = \int_I |f|^2 \geq 0.$$

$\langle f, f \rangle = 0$ iff $f \equiv 0$.

Define the L_2 norm by $\|f\|_2 = \sqrt{\langle f, f \rangle}$. Note $\|f\|_2 = 0$ iff $f \equiv 0$. I claim that $|\langle f, g \rangle| \leq \|f\|_2 \cdot \|g\|_2$.

$$|\langle f, g \rangle| \leq \int_I |f \bar{g}| = \int_I |f| \cdot |g| \leq \sqrt{\int_I |f|^2} \cdot \sqrt{\int_I |g|^2} = \|f\|_2 \cdot \|g\|_2.$$

I claim if $f : I \rightarrow \mathbb{R}$ is bounded continuous, then $|\int_I f| \leq \int_I |f|$. We have

$$\begin{aligned} \left| \int_I f \right| &= \sqrt{\left(\int_I \Re(f) \right)^2 + \left(\int_I \Im(f) \right)^2} \\ &\leq \int_I \sqrt{\Re(f)^2 + \Im(f)^2}. \end{aligned}$$

Let $f : I \rightarrow \mathbb{R}$ be continuous. Given $n \in \mathbb{Z}^+$ and $i \in [[n]]$, let $x_i = i/n$. Then $\int_I f = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(x_i) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f(x_i) = \lim_{n \rightarrow \infty} \frac{1}{2n} \sum_{i=0}^{n-1} (f(x_i) + f(x_{i+1}))$.

For $n \in \mathbb{Z}^+$, define $U_n = \frac{1}{n} \sum_{i=0}^{n-1} \sup f([x_i, x_{i+1}])$ and $L_n = \frac{1}{n} \sum_{i=0}^{n-1} \inf f([x_i, x_{i+1}])$. Since f is uniformly continuous, given $\epsilon > 0$, pick $\delta > 0$ such that $|x - y| < \delta \rightarrow |f(x) - f(y)| < \epsilon$. By the Archimedean principle, pick $N \in \mathbb{Z}^+$ such that

$1/N < \delta$. Then for $n \geq N$, we have for $x, x' \in [x_i, x_{i+1}]$ that $|f(x) - f(x')| < \epsilon$. It follows by EVT

$$U_n - L_n < \frac{1}{n} \sum_{i=0}^{n-1} \epsilon = \epsilon,$$

so $\lim_{n \rightarrow \infty} (U_n - L_n) = 0$. Further,

$$\lim_{n \rightarrow \infty} (U_n - L_n) \geq \lim_{n \rightarrow \infty} (U_n - \int_I f) \geq 0,$$

so by Squeeze lemma, $\lim_{n \rightarrow \infty} U_n = \int_I f$, so $\lim_{n \rightarrow \infty} L_n = \int_I f$ as well.

Observe that

$$L_n \leq \frac{1}{n} \sum_{i=0}^{n-1} f(x_i) \leq U_n,$$

so

$$0 \leq \frac{1}{n} \sum_{i=0}^{n-1} f(x_i) - L_n \leq U_n - L_n,$$

so by Squeeze lemma, $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(x_i) - L_n = 0$. Since $\lim_{n \rightarrow \infty} L_n = \int_I f$, we have $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(x_i) = \int_I f$. A similar result follows for the right Riemann sums. Taking the average gives the Trapezoid sums.

Now, let $f, g : I \rightarrow \mathbb{R}$ be continuous. I claim that $\sqrt{(\int_I f)^2 + (\int_I g)^2} \leq \int_I \sqrt{f^2 + g^2}$.

Fix $n \in \mathbb{Z}^+$. Then by the Triangle Inequality on complex numbers,

$$\begin{aligned} \sqrt{\left(\frac{1}{n} \sum_{i=0}^{n-1} f(x_i)\right)^2 + \left(\frac{1}{n} \sum_{i=0}^{n-1} g(x_i)\right)^2} &= \frac{1}{n} \sqrt{\left(\sum_{i=0}^{n-1} f(x_i)\right)^2 + \left(\sum_{i=0}^{n-1} g(x_i)\right)^2} \\ &\leq \frac{1}{n} \sum_{i=0}^{n-1} \sqrt{f(x_i)^2 + g(x_i)^2}. \end{aligned}$$

Note that $\sqrt{\cdot}$, $\sqrt{f^2 + g^2}$ are continuous, so the limits of the LHS and RHS as $n \rightarrow \infty$ exist. Taking limits,

$$\sqrt{\left(\int_I f\right)^2 + \left(\int_I g\right)^2} \leq \int_I \sqrt{f^2 + g^2}.$$

Hence, if $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$, then $|\int_I f| \leq \int_I |f|$. I claim for all $f, g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$ that $|\langle f, g \rangle| \leq \|f\|_2 \cdot \|g\|_2$.

$$\begin{aligned} |\langle f, g \rangle| &= \left| \int_I f \bar{g} \right| \leq \int_I |f| \cdot |g| \leq \sqrt{\int_I |f|^2} \cdot \sqrt{\int_I |g|^2} \\ &= \|f\|_2 \cdot \|g\|_2. \end{aligned}$$

I claim that $\|f + g\|_2 \leq \|f\|_2 + \|g\|_2$.

Proof:

$$\begin{aligned} \|f + g\|_2^2 &= |\langle f + g, f + g \rangle| = |\langle f, f \rangle + \langle f, g \rangle + \langle g, f \rangle + \langle g, g \rangle| \\ &\leq \langle f, f \rangle + |\langle f, g \rangle| + |\langle g, f \rangle| + \langle g, g \rangle \\ &\leq \|f\|_2^2 + 2\|f\|_2 \cdot \|g\|_2 + \|g\|_2^2 = (\|f\|_2 + \|g\|_2)^2, \end{aligned}$$

so $\|f + g\|_2 \leq \|f\|_2 + \|g\|_2$.

Claim: if $\langle f, g \rangle = 0$, then $\|f + g\|_2^2 = \|f\|_2^2 + \|g\|_2^2$.

Proof:

$$\langle f + g, f + g \rangle = \langle f, f \rangle + \langle f, g \rangle + \langle g, f \rangle + \langle g, g \rangle = \langle f, f \rangle + \langle g, g \rangle.$$

Claim: for all $c \in \mathbb{C}$, we have $\|cf\|_2 = |c| \cdot \|f\|_2$.

Proof:

$$\|cf\|_2^2 = \langle cf, cf \rangle = c\bar{c}\langle f, f \rangle = |c|^2 \cdot \|f\|_2^2,$$

so $\|cf\|_2 = |c| \cdot \|f\|_2$.

Consider $h = \langle f, f \rangle g - \langle g, f \rangle f$.

$$\begin{aligned} \langle h, h \rangle &= \langle f, f \rangle^2 \langle g, g \rangle - \langle g, f \rangle^2 \langle f, f \rangle - \langle g, f \rangle \langle f, g \rangle \langle f, f \rangle + \langle g, f \rangle^2 \langle f, f \rangle \\ &= \langle f, f \rangle \cdot (\langle f, f \rangle \langle g, g \rangle - \langle g, f \rangle \langle f, g \rangle) \geq 0. \end{aligned}$$

If $\langle f, f \rangle = 0$, then $f \equiv 0$, for which the result holds. Else, $\langle f, f \rangle > 0$, so $|\langle f, g \rangle|^2 \leq \|f\|_2^2 \cdot \|g\|_2^2$, so $|\langle f, g \rangle| \leq \|f\|_2 \cdot \|g\|_2$, as desired.

Let $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{R})$. Show that $\|f\|_2 \leq \|f\|_\infty$. We have

$$\langle f, f \rangle = \int_I |f|^2 \leq \int_I \|f\|_\infty^2 = \|f\|_\infty^2,$$

so $\|f\|_2 \leq \|f\|_\infty$.

3 Trigonometric Polynomials

Lemma: $\langle e_n, e_m \rangle = \delta_{n,m}$.

Proof: We have

$$\begin{aligned}\langle e_n, e_m \rangle &= \int_I e^{2\pi i n x} \cdot \overline{e^{2\pi i m x}} dx = \int_I e^{2\pi i (n-m)x} dx \\ &= \int_I \cos 2\pi(n-m)x dx + i \int_I \sin 2\pi(n-m)x dx \\ &= \delta_{n,m}.\end{aligned}$$

In particular, $\|e_n\| = 1$ for all $n \in \mathbb{Z}$.

Lemma: Let $f = \sum_{n=-N}^N c_n e_n$. Then $\langle f, e_m \rangle = c_m$.

Proof:

$$\begin{aligned}\langle f, e_m \rangle &= \left\langle \sum_{n=-N}^N c_n e_n, e_m \right\rangle = \sum_{n=-N}^N \langle c_n e_n, e_m \rangle = \sum_{n=-N}^N c_n \langle e_n, e_m \rangle \\ &= c_m.\end{aligned}$$

Lemma: $\|f\|_2^2 = \sum_{n=-N}^N |c_n|^2$.

Proof:

$$\|f\|_2^2 = \sum_{n=-N}^N \sum_{m=-N}^N c_n \overline{c_m} \langle c_n e_n, e_m \rangle = \sum_{n=-N}^N |c_n|^2,$$

as desired.

4 Periodic Convolutions

Suppose $f, g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Define $f * g(x) = \int_I f(t)g(x-t)dt$.

Closure. Suppose $f, g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Then $f * g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. We compute

$$(f * g)(x+1) = \int_I f(t)g(x+1-t)dt = \int_I f(t)g(x-t)dt = (f * g)(x).$$

Suppose $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Then f is uniformly continuous, ie for all $\epsilon > 0$, there exists $\delta > 0$ such that $|x-y| < \delta \rightarrow |f(x) - f(y)| < \epsilon$.

Proof: Let $\epsilon > 0$. Observe that $f_1 : I \rightarrow \mathbb{R}$ is uniformly continuous since I is compact. Pick $\delta_1 > 0$ such that $|x-y| < \delta_1 \rightarrow |f(x) - f(y)| < \epsilon$. Pick $\delta_2 > 0$ such that $|x-y| < \delta_2 \rightarrow |f(x) - f(y)| < \epsilon$ for all $x, y \in I + 1/2$. Assume

$\delta_1, \delta_2 < 1/2$. Let $\delta = \min(\delta_1, \delta_2)$. Suppose $x, y \in \mathbb{R}$ such that $|x - y| < \delta$. WLOG $x \leq y$. Pick $n \in \mathbb{Z}$ such that $\frac{1}{2}n \leq x < \frac{1}{2}(n+1)$. Then $\frac{1}{2}n \leq y < \frac{1}{2}n + 1$. Then the previous cases give $|f(x) - f(y)| < \epsilon$.

I claim that $f * g$ is uniformly continuous. Let $\epsilon > 0$. Pick $\delta > 0$ such that $|x - y| < \delta \rightarrow |g(x) - g(y)| < \epsilon$. Then

$$\begin{aligned} |f * g(x) - f * g(y)| &= \left| \int_I f(t)(g(x-t) - g(y-t))dt \right| \\ &\leq \int_I |f(t)| \cdot |g(x-t) - g(y-t)|dt \\ &\leq \int_I |f(t)| \cdot \epsilon = \epsilon \int_I |f(t)| = \epsilon C \end{aligned}$$

for some constant C independent of x, y , so $f * g$ is indeed uniformly continuous.

Commutativity.

$$\begin{aligned} (f * g)(x) &= \int_I f(t)g(x-t)dt = - \int_x^{x-1} f(x-t)g(t)dt \\ &= \int_{x-1}^x g(t)f(x-t)dt = \int_{x-1}^0 g(t)f(x-t)dt + \int_0^x g(t)f(x-t)dt \\ &= \int_x^1 g(t-1)f(x+1-t)dt + \int_0^x g(t)f(x-t)dt \\ &= \int_x^1 g(t)f(x-t)dt + \int_0^x g(t)f(x-t)dt \\ &= \int_0^1 g(t)f(x-t)dt = (g * f)(x). \end{aligned}$$

Bilinearity. Since $*$ is commutative, it suffices to show linearity in one argument.

$$\begin{aligned} (f + g) * h(x) &= \int_I (f + g)(t)h(x-t)dt \\ &= \int_I f(t)h(x-t)dt + \int_I g(t)h(x-t)dt \\ &= f * h(x) + g * h(x). \end{aligned}$$

$$cf * h(x) = \int_I cf(t)h(x-t)dt = c \int_I f(t)h(x-t)dt = c(f * h)(x).$$

5 Weierstrass Approximation Theorem

Let $\epsilon > 0$ and $0 < \delta < 1$. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ is continuous and supported on $[-1, 1]$. f is said to be an (ϵ, δ) approximation of the identity if $\int_{-1}^1 f(x)dx = 1$

and $|f(x)| \leq \epsilon$ for $|x| \geq \delta$. Claim: there exists a polynomial (ϵ, δ) approximation to the identity for every ϵ, δ .

Proof: Consider $f(x) = C(1 - x^2)^n$, where $C = (\int_{-1}^1 (1 - x^2)^n dx)^{-1}$. Since $(1 - x^2)^n \not\equiv 0$, so $C > 0$ is well-defined. Note $\int_{-1}^1 f(x) dx = 1$ and f is nonnegative.

Observe that

$$\begin{aligned} C^{-1} &= 2 \int_0^1 (1 - x^2)^n dx = 2 \int_0^{1/\sqrt{n}} (1 - x^2)^n dx + 2 \int_{1/\sqrt{n}}^1 (1 - x^2)^n dx \\ &\geq 2 \int_0^{1/\sqrt{n}} (1 - nx^2) dx = 2(1/\sqrt{n} - \frac{n}{3} \cdot n^{-3/2}) = \frac{4}{3\sqrt{n}}. \end{aligned}$$

Hence, $C \leq \frac{3\sqrt{n}}{4}$. It follows for $|x| \geq \delta$ that $f(x) \leq \frac{3\sqrt{n}}{4}(1 - \delta^2)^n$. Let $0 < a < 1$. We have $(1/a)^n \geq 1 + n(1/a - 1)$, so $a^n \leq \frac{1}{1+n(1/a-1)}$. We compute

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{1 + n(1/a - 1)} = 0,$$

so by Squeeze theorem, $\lim_{n \rightarrow \infty} \sqrt{n}a^n = 0$. It follows there exists $n \in \mathbb{Z}^+$ such that $f(x) \leq \epsilon$ for $|x| \geq \delta$.

Define $F_N = \sum_{n=-N}^N (1 - |n|/N) e_n$. Let $D_n = \sum_{k=-n}^n e_k$. Observe that $\int_I D_n(x) dx = 1$ and $F_N = \frac{1}{N} \sum_{n=0}^{N-1} D_n$. Note that D_N is real-valued since e_0 is real and the conjugate functions e_n, e_{-n} have equal coefficients. We compute when $x \notin \mathbb{Z}$:

$$\begin{aligned} D_n(x) &= \sum_{k=-n}^n (e(x))^k = \frac{e(x)^{2n+1} - 1}{e(x)^n(e(x) - 1)} = \frac{e(x)^{n+1} - e(x)^{-n}}{e(x) - 1} \\ &= \frac{(\cos 2\pi(n+1)x - \cos 2\pi nx) + i(\sin 2\pi(n+1)x + \sin 2\pi nx)}{(\cos 2\pi x - 1) + i \sin 2\pi x}. \end{aligned}$$

In general,

$$\Re\left(\frac{a + bi}{c + di}\right) = \frac{ac + bd}{c^2 + d^2},$$

so

$$\begin{aligned} D_n(x) &= \Re(D_n(x)) = \frac{(\cos 2\pi(n+1)x - \cos 2\pi nx)(\cos 2\pi x - 1) + (\sin 2\pi(n+1)x + \sin 2\pi nx) \sin 2\pi x}{(\cos 2\pi x - 1)^2 + \sin^2 2\pi x} \\ &= \frac{-2 \sin \pi(2n+1)x \sin \pi x \cdot (-2 \sin^2 \pi x) + 2 \sin \pi(2n+1)x \cos \pi x \cdot 2 \cos \pi x \sin \pi x}{2 - 2 \cos 2\pi x} \\ &= \frac{4 \sin \pi(2n+1)x \cdot \sin \pi x (\sin^2 \pi x + \cos^2 \pi x)}{4 \sin^2 \pi x} = \frac{\sin \pi(2n+1)x}{\sin \pi x} \end{aligned}$$

Then

$$\begin{aligned} F_N(x) &= \frac{1}{N} \sum_{n=0}^{N-1} \frac{\sin \pi(2n+1)x}{\sin \pi x} = \frac{1}{2N \sin^2 \pi x} \sum_{n=0}^{N-1} \cos 2\pi n x - \cos 2\pi(n+1)x \\ &= \frac{1 - \cos 2\pi N x}{2N \sin^2 \pi x} = \frac{\sin^2 \pi N x}{N \sin^2 \pi x}. \end{aligned}$$

It follows F_N is nonnegative. We also have the formula

$$F_N(x) = \frac{1 - \cos 2\pi N x}{N(1 - \cos 2\pi x)}.$$

Suppose $\delta \leq x \leq 1 - \delta$ for some $\delta < 1/2$. Then $F_N(x) \leq \frac{1}{N(1 - \cos 2\pi\delta)}$. Note that $1 - 2\cos 2\pi\delta > 0$. Suppose $\epsilon > 0$ is given. By the Archimedean principle, pick $N \in \mathbb{Z}^+$ such that $\frac{1}{N(1 - \cos 2\pi\delta)} < \epsilon$.

Suppose F_N is an (ϵ, δ) approximation to the identity and $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Then I can bound $\|f * F_N - f\|_\infty$.

Proof: Let f be bounded by $M > 0$. Let $\epsilon > 0$. Since f is uniformly continuous, pick $1/2 > \delta > 0$ such that $|x - y| < \delta \rightarrow |f(x) - f(y)| < \epsilon$. Pick $N \in \mathbb{Z}^+$ such that F_N is an (ϵ, δ) approximation to the identity. Note that

$$\begin{aligned} \int_0^\delta F_N(x) dx + \int_{1-\delta}^1 F_N(x) dx &= \int_0^1 F_N(x) dx - \int_\delta^{1-\delta} F_N(x) dx \\ &\geq 1 - \epsilon(1 - 2\delta) \geq 1 - \epsilon. \end{aligned}$$

Let $x \in I$. We have

$$f * F_N(x) = \int_0^\delta F_N(t) f(x-t) dt + \int_\delta^{1-\delta} F_N(t) f(x-t) dt + \int_{1-\delta}^1 F_N(t) f(x-t) dt.$$

Note

$$\begin{aligned} &\left| \int_0^\delta F_N(t) f(x-t) dt + \int_{1-\delta}^1 F_N(t) f(x-t) dt - \int_0^\delta F_N(t) f(x) dt - \int_{1-\delta}^1 F_N(t) f(x) dt \right| \\ &\leq \left| \int_0^\delta F_N(t) (f(x-t) - f(x)) dt \right| + \left| \int_{1-\delta}^1 F_N(t) (f(x-t) - f(x)) dt \right| \\ &\leq \int_0^\delta F_N(t) \cdot |f(x-t) - f(x)| dt + \int_{1-\delta}^1 F_N(t) \cdot |f(x-t) - f(x)| dt \\ &\leq \epsilon \cdot \left(\int_0^\delta F_N(t) dt + \int_{1-\delta}^1 F_N(t) dt \right) \leq \epsilon(1 - \epsilon) \leq \epsilon. \end{aligned}$$

And

$$\left| \int_0^\delta F_N(t) f(x) dt + \int_{1-\delta}^1 F_N(t) f(x) dt - f(x) \right| \leq M \cdot \left| \int_0^\delta F_N(t) dt + \int_{1-\delta}^1 F_N(t) dt - 1 \right| \leq M\epsilon.$$

And

$$\left| \int_{\delta}^{1-\delta} F_N(t) f(x-t) dt \right| \leq \int_{\delta}^{1-\delta} F_N(t) \cdot |f(x-t)| dt \leq M \int_{\delta}^{1-\delta} F_N(t) dt \leq M\epsilon.$$

It follows by Triangle inequality that $|f * F_N(x) - f(x)| \leq \epsilon(2M + 1)$ for all $x \in I$. Note that M is constant and $f * F_N$ is a trigonometric polynomial, AWD.

6 The Fourier and Plancherel Theorems

For any $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$, the series $\sum_{n=-N}^N \hat{f}(n)e_n$ converges to f in the L^2 metric.

Pick P such that $\|P - f\|_{\infty} \leq \epsilon$, where $P = \sum_{n=-N_0}^{N_0} c_n e_n$. Let $N \geq N_0$. Define $F_N = \sum_{n=-N}^N \hat{f}(n)e_n$. Observe that $\langle f - F_N, e_m \rangle = 0$ for $|m| \leq N$. It follows $\langle f - f_N, f_N - P \rangle = 0$. Then

$$\begin{aligned} \langle f - f_N, f - f_N \rangle &= \langle f - f_N, f - f_N \rangle + \langle f - f_N, f_N - P \rangle \\ &= \langle f - f_N, f - P \rangle \leq \|f - f_N\|_2 \cdot \|f - P\|_2 \leq \|f - f_N\|_2 \cdot \|f - P\|_{\infty}, \end{aligned}$$

so

$$\|f - f_N\|_2^2 \leq \|f - f_N\|_2 \cdot \epsilon,$$

so either $\|f - f_N\|_2 = 0$ or $\|f - f_N\|_2 \leq \epsilon$, so $\|f - F_N\|_2 \leq \epsilon$ either way.

Let $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$ and suppose $\sum_{n=-\infty}^{\infty} |\hat{f}(n)|$ converges absolutely. Then the sequence (F_N) converges absolutely to f .

By Weierstrass M -test, the functions (F_N) converge uniformly to $g = \sum_{n=-\infty}^{\infty} \hat{f}(n)e_n$. Since each $F_N \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$, $g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$ as well. It follows that the (F_N) converge in the L^2 metric to g as well. The previous result shows that the (F_N) converge to f in the L^2 metric, so by Uniqueness of limits, $f \equiv g$.

For any $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$, the series $\sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2$ is absolutely convergent and $\|f\|_2^2 = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2$.

Proof: Suppose that $f_N \rightarrow f$ in L^2 . Then I claim that $\|f_N\|_2 \rightarrow \|f\|_2$ in $\mathbb{R}$. This follows from the inequality

$$|\|f\|_2 - \|f_N\|_2| \leq \|f - f_N\|_2.$$

Hence, $\|f_N\|_2^2 \rightarrow \|f\|_2^2$ in $\mathbb{R}$. Since $\|f_N\|_2^2 = \sum_{n=-N}^N |\hat{f}(n)|^2$, we have

$$\lim_{N \rightarrow \infty} \sum_{n=-N}^N |\hat{f}(n)|^2 = \|f\|_2^2.$$

Since the terms of the sum in the LHS are nonnegative, the LHS is absolutely convergent.

Let P be a tpoly and $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Then $\widehat{f * P}(n) = \hat{f}(n)\hat{P}(n)$.

Proof: Write $P = \sum_{k=-N}^N c_k e_k$. Then

$$\widehat{f * P} = \sum_{k=-N}^N c_k \cdot \widehat{f * e_k} = \sum_{k=-N}^N c_k \hat{f}(k) e_k,$$

so

$$\widehat{f * P}(n) = \begin{cases} c_n \hat{f}(n) & \text{if } |n| \leq N, \\ 0 & \text{if } |n| > N. \end{cases} = \hat{f}(n)\hat{P}(n).$$

Let $f \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Suppose that f' is continuous. Then $f' \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$ and $\hat{f}'(n) = 2\pi i n \hat{f}(n)$.

Proof: One can show $f'(x+1) = f'(x)$ for all $x \in \mathbb{R}$ by periodicity of f . By Integration by parts,

$$\begin{aligned} \hat{f}'(n) &= \int_I f'(x) e_{-n}(x) dx = \left(f(x) e_{-n}(x) - \int f(x) (-2\pi i n e_{-n}(x)) dx \right)_I \\ &= 2\pi i n \int_I f(x) e_{-n}(x) dx = 2\pi i n \hat{f}(n). \end{aligned}$$

Let $f, g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$. Prove Parseval's identity:

$$\int_I f(x) \overline{g(x)} dx = \sum_{n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{g}(n)}.$$

Observe by Parseval's theorem,

$$\begin{aligned} \|f + g\|_2^2 &= \sum_{n \in \mathbb{Z}} |\hat{f}(n) + \hat{g}(n)|^2 = \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 + \hat{f}(n) \overline{\hat{g}(n)} + \overline{\hat{f}(n)} \hat{g}(n) + |\hat{g}(n)|^2, \\ \|f - g\|_2^2 &= \sum_{n \in \mathbb{Z}} |\hat{f}(n) - \hat{g}(n)|^2 = \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 - \hat{f}(n) \overline{\hat{g}(n)} - \overline{\hat{f}(n)} \hat{g}(n) + |\hat{g}(n)|^2. \end{aligned}$$

Subtracting the LHS's, we get

$$\langle f + g, f + g \rangle - \langle f - g, f - g \rangle = 2(\langle f, g \rangle + \langle g, f \rangle) = 4\Re \int_I f(x) \overline{g(x)} dx.$$

Subtracting the RHS's, we get

$$2 \sum_{n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{g}(n)} + \overline{\hat{f}(n)} \hat{g}(n) = 4 \sum_{n \in \mathbb{Z}} \Re(\hat{f}(n) \overline{\hat{g}(n)}) = 4\Re \sum_{n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{g}(n)}.$$

Hence,

$$\Re \int_I f(x) \overline{g(x)} dx = \Re \sum_{n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{g}(n)}.$$

Apply the result to $-if, g$, so

$$\Im \int_I f(x) \overline{g(x)} dx = \Im \sum_{n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{g}(n)},$$

so the result follows.

General Fourier series. Let $f \in C(\mathbb{R}/L\mathbb{Z} : \mathbb{C})$. Define

$$c_n = \frac{1}{L} \int_{[0, L]} f(x) e^{-2\pi i n x / L} dx$$

Consider $g \in C(\mathbb{R}/\mathbb{Z} : \mathbb{C})$ given by $g(x) = f(Lx)$. The previous results show that $g_N = \sum_{n=-N}^N \hat{g}(n) e_n$ converges in the L^2 metric to g . Note that $\hat{g}(n) = \int_I g(x) e^{-2\pi i n x} dx = \frac{1}{L} \int_{[0, L]} f(y) e^{-2\pi i n y / L} dy = c_n$. Given $\epsilon > 0$, pick $N_0 \in \mathbb{Z}^+$ such that $N \geq N_0 \rightarrow \|g_N - f\|_2 \leq \epsilon/L$, ie

$$\begin{aligned} \int_I |g_N(x) - g(x)|^2 dx &= \frac{1}{L} \int_I |g_N(y/L) - f(y)|^2 dy \\ &= \frac{1}{L} \int_{[0, L]} \left| \sum_{n=-N}^N c_n e^{-2\pi i n y / L} - f(y) \right|^2 dy \leq \epsilon/L, \end{aligned}$$

implying the result.

Suppose the series $\sum_{n \in \mathbb{Z}} |c_n|$ is absolutely convergent. Then $\sum_{n \in \mathbb{Z}} |\hat{g}(n)|$ is absolutely convergent, so the g_N converge uniformly to g , so the $\sum_{n=-\infty}^{\infty} c_n e^{-2\pi i n x / L}$ converge uniformly to f .

General Plancherel's theorem.

$$\|g\|_2^2 = \int_I |g(x)|^2 dx = \frac{1}{L} \int_I |f(y)|^2 dy$$

and $\sum_{n \in \mathbb{Z}} |\hat{g}(n)|^2 = \sum_{n \in \mathbb{Z}} |c_n|^2$.